1 Measurement with the Multimeter

1.1 Internal resistance of the Voltmeter

Expected value

$$U = RI \Rightarrow I = \frac{U_0}{R_1 + R_2} \tag{1.1}$$

$$U_{\text{out}} = R_2 I \tag{1.2}$$

$$\Rightarrow U_{\text{out}} = U_0 \cdot \frac{R_2}{R_2 + R_1} \tag{1.3}$$

With internal resitor

$$\frac{1}{R_i} = \frac{1}{R_E} - \frac{1}{R_2} \tag{1.4}$$

$$U_0 = I \cdot R_{\text{tot}} \tag{1.5}$$

$$I = \frac{U_0}{R_1 + R_E} = \frac{U_{\text{out}}}{R_E} \tag{1.6}$$

$$\Rightarrow R_i = \frac{R_1 R_2 U_{\text{out}}}{R_2(U_0 - U_{\text{out}}) - R_1 U_{\text{out}}} \tag{1.7}$$

1.2 Compensation Circuit

1.3 Internal resistance of the Amperemeter

$$R_E = R_i + R_3 \tag{1.8}$$

$$U_0 = I\left(R_1 + \frac{R_E R_2}{R_E + R_2}\right) \tag{1.9}$$

$$I = I_{\text{meas}}\left(1 + \frac{R_E}{R_2}\right) \tag{1.10}$$

$$\Rightarrow R_i = \frac{\frac{R_2 U_0}{I_{\text{meas}}} - R_1 R_2}{R_1 + R_2} - R_3 \tag{1.11}$$

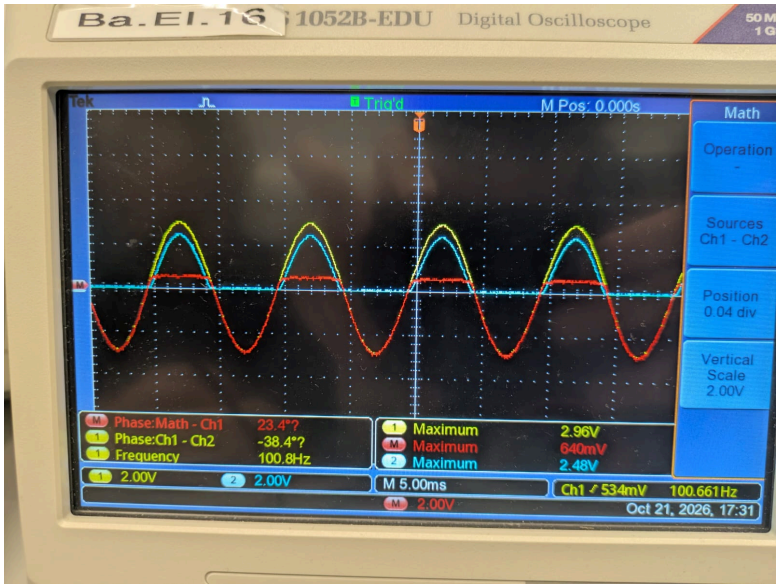
1.4 Four probe Measurement

2 Measurement woth the Oscilloscope

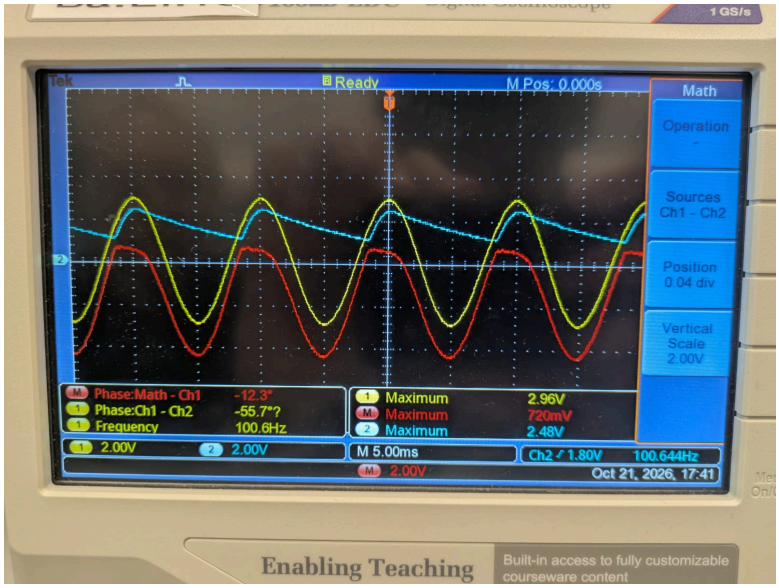
2.1

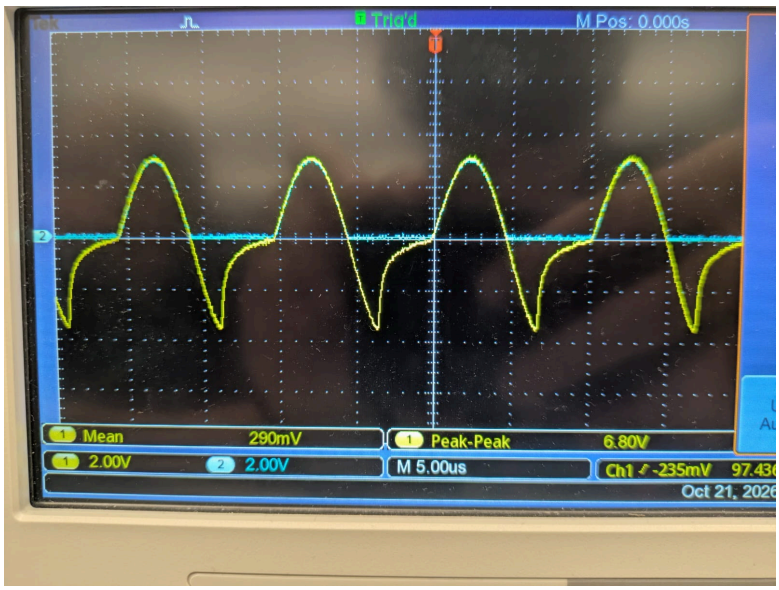
2.2

2.3

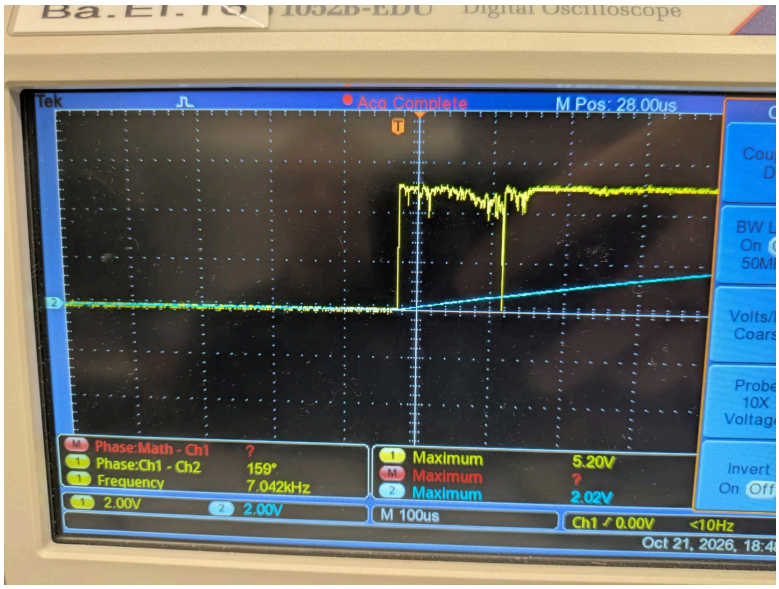


2.4




Trig'd
M Pos: 0.000s
Mean 290mV
Peak-Peak 6.80V
2.00V 2.00V
M 5.00us
Ch1 -235mV
Oct 21, 2026

2.5


Tek
Acq Complete
M Pos: 28.00us
Phase:Math - Ch1 ?
Phase:Ch1 - Ch2 159°
Frequency 7.042kHz
Maximum 5.20V
Maximum ?
Maximum 2.02V
2.00V 2.00V
M 100us
Ch1 0.00V
<10Hz
Oct 21, 2026